

AHHYUN YUH

<https://lucy0828.github.io/ahhyunyuh.github.io/#>

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EDUCATION

Sep 2022 - Present	University of Pennsylvania <i>PhD Candidate in Computer and Information Science</i>	Philadelphia, PA, USA
Mar 2020 - Feb 2022	Kyungpook National University <i>Master of Science in Electronic Engineering</i>	Daegu, Korea
Mar 2015 - Feb 2020	Kyungpook National University <i>Bachelor of Science in Electronic Engineering</i>	Daegu, Korea
Feb 2018 - Aug 2018	University of Ljubljana <i>Exchange Student Program</i>	Ljubljana, Slovenia

PROJECTS

iCareChat: Voice-Based LLM Agent for Older Adults (Ongoing, Lead Developer)

Designed and currently leading the development of iCareChat, a privacy-preserving, HIPAA-compliant, voice-based chatbot application using large language models. Built with a multi-agent framework and real-time audio processing pipeline to deliver empathetic, accessible support for older adults. Evaluated via IRB-approved clinical studies across the US and Japan to assess mental health outcomes and usability.

TrueDepth 3D Face Reconstruction for Graves' Disease Monitoring (Ongoing)

Developing an iOS-based facial monitoring tool using Apple's ARKit and TrueDepth camera to track morphological changes in patients with Graves' disease. Combines RGB-D alignment, landmark detection, and privacy-focused data handling to enable in-home monitoring integrated into clinical workflows.

iCareLoop: Sensing Framework for Detecting Social Isolation and Loneliness

Led the end-to-end design and implementation of iCareLoop, an IoT-based, passive monitoring system to identify social isolation and loneliness in older adults through behavioral signal analysis. Applied time-series modeling and pattern recognition to sensor data from US and Japanese eldercare settings. System validated in multi-country deployment studies and published across major health-tech venues.

BlinkWise: mmWave-Based Mental State Monitoring via Blink Dynamics

Co-developed BlinkWise, a low-power wearable device using mmWave radar and ESP32 microcontroller to monitor blink patterns and infer real-time cognitive states. Designed custom signal processing pipeline (FFT, beamforming, adaptive filtering) for high-accuracy, low-latency blink detection. System supports applications in mental workload tracking and user-context inference.

Real-Time Sound Event Classification for Human Activity Recognition

Developed an edge-based ambient sound classification system for real-time, privacy-conscious human activity recognition in smart homes. Designed lightweight audio feature extractors and trained a CNN-based classifier deployable on resource-constrained embedded devices. Achieved robust performance for daily living activity monitoring with no camera-based intrusion.

PUBLICATIONS

1. Teruhiro Mizumoto, Viktor Erdélyi, Xiayan Ji, Ahhyun Yuh, Tekeshi Nakagawa, Takashi Suehiro, Hajime Nagahara, Yasuyuki Gondo, George Demiris, Oleg Sokolsky, Insup Lee, Teruo Higashino. IoT-Based Loneliness Monitoring for Older Adults: Lessons from Japanese and US Communities. *IEEE Access*, 2025. [Submitted]
2. Xiayan Ji, Ahhyun Yuh, Viktor Erdélyi, Teruhiro Mizumoto, Hyonyoung Choi, Sean Lee Harrison, Emma Cho, Takashi Suehiro, Takeshi Nakagawa, Cheng Yutian, Yasuyuki Gondo, Hajime Nagahara, Teruo Higashino, George Demiris, Oleg Sokolsky, Insup Lee. SMILE: Sensor Data-Driven Detection and Proximate Cause Analysis for Loneliness in Older Adults. *ACM Conference on Health, Inference, and Learning (ACM Health)*, 2025. [Submitted]
3. Dongyin Hu, Xin Yang, Ahhyun Yuh, Zihao Wang, Lama A. Al-Aswad, Insup Lee, Mingmin Zhao. Tracking Blink Dynamics and Mental States on Glasses. *Proceedings of the ACM on Mobile Systems, Applications, and Services (MobiSys)*, 2025. [Accepted]
4. Emma Cho, Hannah Cho, George Demiris, Sean Harrison, Xiayan Ji, Ahhyun Yuh, Oleg Sokolsky, Insup Lee. Older Adults' Perceptions and Attitudes Toward Passive Sensors to Measure Loneliness: A Qualitative Study. *SAGE Open Aging*, 2025.
5. Emma Cho, George Demiris, Hannah Cho, Insup Lee, Ahhyun Yuh, Xiayan Ji, Sean Lee Harrison, Oleg Sokolsky. Perceptions and Attitudes Toward Loneliness Detection Using Passive Sensing Technologies in Older Adults. *Innovation in Aging*, Vol. 8, Suppl. 1, 2024.
6. Cheolhwan Lee, Ahhyun Yuh, Soonju Kang. Real-Time Prediction of Resident ADL Using Edge-Based Time-Series Ambient Sound Recognition. *Sensors*, Vol. 24, No. 19, 2024.
7. Xiayan Ji, Ahhyun Yuh, Viktor Erdélyi, Teruhiro Mizumoto, Hyonyoung Choi, Sean Lee Harrison, Emma Cho, James Weimer, Hajime Nagahara, Teruo Higashino, George Demiris, Oleg Sokolsky, Insup Lee. Exploring Effective Sensing Indicators of Loneliness for Elderly Community in US and Japan. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*, 2024. <https://doi.org/10.1145/3613905.3637123>
8. Xiayan Ji, Ahhyun Yuh, Hyonyoung Choi, Insup Lee, Viktor Erdélyi, Teruhiro Mizumoto, Sean Lee Harrison, James Weimer, George Demiris, Takeshi Nakagawa, Takashi Suehiro, Yasuyuki Gondo, Hajime Nagahara, Teruo Higashino, Oleg Sokolsky. iCareLoop: Data Management System for Monitoring Gerontological Social Isolation and Loneliness. *Proceedings of the 14th International Conference on Mobile Computing and Ubiquitous Networking (ICMU)*, 2023.
9. Xiayan Ji, Xian Li, Ahhyun Yuh, Amanda Watson, Claire Kendell, James Weimer, Hajime Nagahara, Teruo Higashino, Teruhiro Mizumoto, Viktor Erdélyi, George Demiris, Oleg Sokolsky, Insup Lee. Short: Integrated Sensing Platform for Detecting Social Isolation and Loneliness in the Elderly Community. *IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 2023. <https://doi.org/10.1145/3580252.3586983>
10. Xiayan Ji, Ahhyun Yuh, Hyonyoung Choi, Amanda Watson, Claire Kendell, Xian Li, James Weimer, Hajime Nagahara, Teruo Higashino, Teruhiro Mizumoto, Viktor Erdélyi, George Demiris, Oleg Sokolsky, Insup Lee. iCareLoop: Closed-Loop Sensing and Intervention for Gerontological Social Isolation and Loneliness. *Proceedings of the ACM/IEEE 14th International Conference on Cyber-Physical Systems (ICCPs)*, 2023.
11. Ahhyun Yuh, Soonju Kang. Real-Time Sound Event Classification for Human Activity of Daily Living Using Deep Neural Network. *Proceedings of the 14th IEEE International Conference on Internet of Things (iThings)*, Melbourne, 2021.
12. Ahhyun Yuh, Soonju Kang. Self-Organized Sensor Fusion Between IoT Devices for Resident Activity Recognition and Evaluation System. *Proceedings of the Institute of Embedded Engineering of Korea Conference*, 2020.

WORK EXPERIENCE

May 2020 - Feb 2022	MATLAB Student Ambassador https://www.facebook.com/groups/KNUMATLAB ▪ Led technical workshops and hands-on MATLAB/Simulink sessions, driving adoption and proficiency among engineering students. ▪ Developed social media strategy and content for the Kyungpook National University MATLAB community, increasing engagement across Facebook and Instagram. ▪ Acted as a technical mentor and liaison between students and MathWorks, promoting MATLAB in curriculum and extracurricular projects. ▪ Strengthened communication and presentation skills by teaching complex software tools to diverse student groups.	MathWorks Korea
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Mar 2020 - Feb 2022	Graduate Research Assistant – Center for Self-Organizing Software http://www.csosp.org/	Kyungpook National University
	- Contributed to the Self-Organized Ambient Assisted Living (SAAL) initiative, designing intelligent IoT systems to enhance in-home care. - Implemented sensor fusion frameworks and adaptive system control techniques for real-world deployment in smart environments. - Collaborated with faculty and engineers to translate academic prototypes into functional, testable solutions for eldercare. - Gained experience in system architecture, data handling, and cross-disciplinary research in embedded software applications.	
Sep 2019 - Feb 2022	Undergraduate Researcher – Real-Time Systems Lab https://sites.google.com/view/knu-rtlab/	Kyungpook National University
	- Designed and implemented real-time audio classification systems for activity recognition in smart home environments. - Conducted performance evaluations and co-authored academic papers, contributing to award-winning research recognized at KNU-EE Congress. - Built expertise in embedded systems, machine learning model deployment, and low-latency signal processing. - Demonstrated strong research-to-product thinking by developing end-to-end solutions suitable for aging-in-place technologies.	
Mar 2018 - Feb 2019	Technical Intern – RLS Merilna Tehnika (Industrial Sensors) https://www.rls.si/	Vodice, Slovenia
	- Supported international product marketing and localization strategy for the Korean market. - Translated technical documentation (e.g., datasheets, product manuals, web content) from English to Korean with high technical accuracy. - Assisted in preparing for SPS Germany 2018, contributing to RLS's international outreach and engineering demonstrations. - Developed cross-cultural communication and global teamwork skills by working across engineering and business teams in Europe and Asia.	

AWARDS AND HONORS

2nd Place, KNU-EE Research Congress – Kyungpook National University, 2021

Recognized for research on real-time sound event classification for activity recognition using deep neural networks.

2nd Place, Startup Idea & Pitching Competition – Gyeongsangbuk-do Creative Content Agency, 2019

Awarded for proposing an innovative e-commerce business model integrating virtual fabric printing and supply chain optimization.

1st Place, Intelligent Robot Tournament – Kyungpook National University, 2015

Designed and programmed a pick-and-place robot to complete dynamic tasks in a competitive environment.